

Weather Station One Part 4 - the Rain Gauge

Harry

VIEW IN BROWSER

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Summary

This is part 4 of a complete and modular weather station project. This is the rain gauge part.

Gadgets > Other Gadgets

Tags: sensor temperature humidity wind weather rain stevensonscreen pressure vane anemometer barometric

Overview on project

This is part of the Weather Station One project, a complete and modular weather station. The design of the system has been driven by a number of requirements:

- collect all common weather channels including temperature, humidity, barometric pressure, rain, wind speed and direction, illumination / sun hours, ground humidity
- modular design allowing one to implement part of it, or extend the range of sensors
- high depth of production; DIY as much as possible
- long range transmission support - with the point of measurement far from home
- integration into a home automation system

- functional, easy to print, and beautiful 3D design

When considering a project like this, please be aware that even with all designs published, it will be some work to do. This includes not only 3D printing, but carpentry work, soldering / electronics, and software in addition.

The parts published are made up from:

1. **Part 1: Central Station.** This is the part all sensors are connected to. It is battery / solar powered to allow placing it in the best place to measure data. It sends data collected using a HC-12 / 433Mhz band (up to 1 kilometer distance) chip.
2. **Part 2: Base Station.** This is the Central Station's counter part. It receives data using another HC-12 chip and is dispatching it to our home-automation system. In addition, it hosts a web server allowing direct access to weather data.
3. **Part 3: Temperature** (and other) **sensor**. This sensor is designed to sit in a proper Stevenson Screen / Wetterhütte. The current version measures ambient temperature, humidity, and barometric pressure using a digital sensor. It is connected to the Central Station.
4. **Part 4: Rain Gauge.** This is a rain gauge collecting rain and feeding it into commuting buckets. The number of commutes is measured using a reed contact triggered a bucket-moved magnet. It is connected to the Central Station.
5. **Part 5: Wind Vane and Anemometer.** Probably the most complex printing part measuring 16 wind directions and wind speed. Similar to the rain gauge, it is a mechanical part. Wind directions are measured using either 8 reed contacts, or an magnetic angle measurement sensor. Wind speed is measured by a single reed contact closed every time the anemometer wings do a full rotation. It is connected to the Central Station.
6. **Part 6: Stevenson Screen / Wetterhütte.** This part is not detailed further because we have built it based on a tutorial published by DWD (Deutscher Wetterdienst). As the tutorial has been removed from their site some time ago, you can find it as PDF file in **Part 1: Central Station.**
7. **Part 7: optional Battery Pack.** The solar module and charger chosen come with a mount for a 14500 sized Lithium battery. Depending on the power consumption of the Central Station (and sensors) you may want to use an external Battery Pack.
8. **Part 8: Software and Electronics.** This part is published on Github.
9. **Part 9: Solar Panel Mount**. This is a solar panel mount as used currently. It allows adjusting the elevation to optimize generation for Winter and Summer seasons. Mid term it will be replaced by a solar tracker following the sun azimuth.

For your convenience, I have created a WeatherStationOne collection including all but parts 6 and 8.

End of overview common to all parts.

Part 4 - Rain Gauge

Update v2:

- **RainGaugeBase** has been modified to feature a much wider gutter.
- Gutter is running downwards now.
- The gap between **Bucket** and **RainGaugeBase** has been increased to work around surface tension from water sitting between bucket and base.
- The gutter is extending beyond the body so water is not running down the body.
- The **Bucket** received a small slant in the top. This works around situation where the bucket is in horizontal position and water collected is distributed evenly to left and right. In case you still observe a horizontal state, consider using a smaller magnet or increase the distance between magnet and reed contact.
- Files are updated and description is adjusted to v2.
- Items to re-print when updating: only **Bucket** and **RainGaugeBase**. Four M3x10 screws need to be replaced by M3x20 screws.
- Two test videos are available here: bucket test, stress test

The rain gauge collects rain for a certain circular area. The water collected is dropping into a bucket that will commute once it is filled. This movement will trigger a reed contact and the ESP32 in the Central Station will count the number of contacts made. From bucket size, the area rain is collected, and the number of counts, we can derive the amount of rain in mm or liters per square meter etc.

As most parts of Weather Station One, we use PETG filament. To be in line with the wind vane / anemometer optically, the rain gauge is coming with sealing to be printed in TPE or TPU.

The box will get wet inside and outside. For all metal parts, please chose stainless steel.

The part requires calibration once it is assembled. You will need to measure the amount of water you can drop into the gauge before the bucket flips. Using a small syringe is convenient.

In case the reed does not trigger when the bucket flips, try to change the distance between reed and magnet using the two screws going through the axle. Using a stronger magnet should be last option because it may add additional resistance for the buckets to move.

Take care with the reed contacts using a glass housing. They break very fast when bending the wires. I glue them into the RainGaugeBase with a two component glue to protect them. The cable going down to the station needs to be clamped to the base to protect the reed wires.

Printing configuration

- Layer height 0.2mm
- 0.4mm nozzle
- 15% infill
- White PETG filament
- Black TPE / TPU filament
- place gauge and base part upside down
- automatic support (from building plate only)

Other parts needed

- 2x miniature steel bearing 3x8x4mm
- 2x M3x10 screws as axle through bearings
- 1x reed contact 2x14mm
- 3x M3x10 screws (RainGaugeBase/RainGaugeBaseBottom)
- 4x M3x20 (M3x16 or 18 is fine too) screws (Gauge/RainGaugeBase)
- 1x magnet 5x1mm (buy some more for fine tuning)
- cable to run from gauge to central station, outer diameter 4.7mm
- 1x MiniDIN (PS/2) 6 pin plug
- 2x M6 nut for the base and as counter nut
- 1x M6 threaded rod as mount to the Stevenson Screen

Closing

This is another free 3D model made available for you. Designing, testing, preparing, uploading and maintaining my models is effort. In case you enjoy them, please send a donation using the coin button in the top right. Thanks a lot!

This remix is based on

Model files



raingaugebaseseal.stl



cableholder.stl



rainseals.3mf

PrusaSlicer file for all seals



raingageseal.stl



raingaugebasebottom.stl



cableseal.stl



gauge.stl



raingauge.3mf



bucket.stl



raingaugebase.stl



raingaugebase_custom.stl

See comment by @aelenkov

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